

Exercise 7

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Problem 1. Let α and β be a complex number. Define $(a_n)_{n \geq 0}$ be a power series expansion.

$$\frac{1}{(1 - \alpha X)(1 - \beta X)} = \sum_{n \geq 0} a_n X^n$$

$$\begin{aligned} \sum_{n \geq 0} a_n X^n &= \frac{1}{(1 - \alpha X)(1 - \beta X)} \\ &= \frac{1}{1 - \alpha X} \frac{1}{1 - \frac{\beta}{\alpha}} + \frac{1}{1 - \beta X} \frac{1}{1 - \frac{\alpha}{\beta}} \\ &= \frac{1}{1 - \alpha X} \frac{\alpha}{\alpha - \beta} - \frac{1}{1 - \beta X} \frac{\beta}{\alpha - \beta} \\ &= \frac{1}{\alpha - \beta} \left(\frac{\alpha}{1 - \alpha X} - \frac{\beta}{1 - \beta X} \right) \\ &= \frac{1}{\alpha - \beta} \left(\sum_{n \geq 0} \alpha^{n+1} X^n - \sum_{n \geq 0} \beta^{n+1} X^n \right) \end{aligned}$$

Thus

$$a_n = \frac{1}{\alpha - \beta} (\alpha^{n+1} - \beta^{n+1})$$

Claim 1a.

$$\sum_{n \geq 0} a_n^2 X^n = \frac{1 + \alpha\beta X}{(1 - \alpha^2 X)(1 - \alpha\beta X)(1 - \beta^2 X)}$$

Proof.

$$\begin{aligned} a_n^2 &= \frac{1}{(\alpha - \beta)^2} (\alpha^{n+1} - \beta^{n+1})^2 \\ &= \frac{1}{(\alpha - \beta)^2} (\alpha^{2(n+1)} - 2\alpha^{n+1}\beta^{n+1} + \beta^{2(n+1)}) \end{aligned}$$

Thus,

$$\sum_{n \geq 0} a_n^2 X^n = \frac{1}{(\alpha - \beta)^2} \left(\frac{\alpha^2}{1 - \alpha^2 X} - \frac{2\alpha\beta}{1 - \alpha\beta X} + \frac{\beta^2}{1 - \beta^2 X} \right)$$

We first calculate the numerator:

$$\begin{aligned}
& \alpha^2(1 - \alpha\beta X)(1 - \beta^2 X) - 2\alpha\beta(1 - \alpha^2 X)(1 - \beta^2 X) \\
& + \beta^2(1 - \alpha^2 X)(1 - \alpha\beta X) \\
& = (\alpha - \beta)^2(1 + \alpha\beta X)
\end{aligned}$$

Thus

$$\begin{aligned}
\sum_{n \geq 0} a_n^2 X^n &= \frac{1}{(\alpha - \beta)^2} \frac{(1 + \alpha\beta X)(\alpha - \beta)^2}{(1 - \alpha^2 X)(1 - \alpha\beta X)(1 - \beta^2 X)} \\
&= \frac{1 + \alpha\beta X}{(1 - \alpha^2 X)(1 - \alpha\beta X)(1 - \beta^2 X)}
\end{aligned}$$

□

Claim 1b.

$$\sum_{n \geq 0} a_{2n} X^n = \frac{\alpha + \beta}{(1 - \alpha^2 X)(1 - \beta^2 X)}$$

Proof.

$$a_{2n} = \frac{1}{\alpha - \beta} (\alpha^{2n+1} - \beta^{2n+1})$$

$$\begin{aligned}
\sum_{n \geq 0} a_{2n} X^n &= \frac{1}{\alpha - \beta} \left(\frac{\alpha}{1 - \alpha^2 X} - \frac{\beta}{1 - \beta^2 X} \right) \\
&= \frac{1}{\alpha - \beta} \left(\frac{\alpha(1 - \beta^2 X) - \beta(1 - \alpha^2 X)}{(1 - \alpha^2 X)(1 - \beta^2 X)} \right) \\
&= \frac{1}{\alpha - \beta} \left(\frac{\alpha - \alpha\beta^2 X - \beta + \alpha^2 \beta X}{(1 - \alpha^2 X)(1 - \beta^2 X)} \right) \\
&= \frac{1}{\alpha - \beta} \left(\frac{\alpha - \beta + \alpha\beta X(\alpha - \beta)}{(1 - \alpha^2 X)(1 - \beta^2 X)} \right) \\
&= \frac{1}{\alpha - \beta} \left(\frac{(\alpha - \beta)(1 + \alpha\beta X)}{(1 - \alpha^2 X)(1 - \beta^2 X)} \right) \\
&= \frac{1 + \alpha\beta X}{(1 - \alpha^2 X)(1 - \beta^2 X)}
\end{aligned}$$

□

Claim 1c.

$$\sum_{n \geq 0} \frac{d(n^2)}{n^s} = \frac{\zeta(s)^3}{\zeta(2s)}$$

Proof. $\forall n \in \mathbb{N}$, for each prime p , $\exists k_p \in \mathbb{N} \cup \{0\}$ such that $n = \prod_p p^{k_p}$ by prime factorization. Thus $n^2 = \prod_p p^{2k_p}$. Thus $d(n) = \prod_p (k_p + 1)$ and $d(n^2) = \prod_p (2k_p + 1)$. Thus we can break the dirichlet series into product of sums.

$$\begin{aligned} \sum_{n \geq 0} \frac{d(n^2)}{n^s} &= \sum_{n \geq 0} \prod_p \frac{(2k_p + 1)}{p^s} \\ &= \prod_p \sum_{k \geq 0} \frac{2k + 1}{p^{ks}} \end{aligned}$$

Lets look at

$$f(x) = \sum_{k \geq 0} (2k + 1)x^k = 2 \sum_{k \geq 0} kx^k + \sum_{k \geq 0} x^k$$

Note for $|x| < 1$

$$\sum_{k \geq 0} x^k = \frac{1}{1 - x}$$

$$\begin{aligned} \sum_{k \geq 0} kx^k &= x \frac{d}{dx} \sum_{k \geq 0} x^k = x \frac{d}{dx} \frac{1}{1 - x} \\ &= \frac{x}{(1 - x)^2} \end{aligned}$$

Thus for $|x| < 1$,

$$\begin{aligned} f(x) &= \frac{2x + (1 - x)}{(1 - x)^2} \\ &= \frac{x + 1}{(1 - x)^2} \end{aligned}$$

Thus,

$$\begin{aligned}
\sum_{n \geq 0} \frac{d(n^2)}{n^s} &= \prod_p \sum_{k \geq 0} \frac{2k+1}{p^{ks}} \\
&= \prod_p f\left(\frac{1}{p^s}\right) \\
&= \prod_p \frac{p^{-s} + 1}{(1 - p^{-s})^2} \\
&= \prod_p \frac{(p^{-s} + 1)(1 - p^{-s})}{(1 - p^{-s})^3} \\
&= \prod_p \frac{(1 - p^{-2s})}{(1 - p^{-s})^3} \\
&= \left(\prod_p (1 - p^{-s})^{-1} \right)^3 \left(\prod_p (1 - p^{-2s})^{-1} \right)^{-1} \\
&= \frac{\zeta(s)^3}{\zeta(2s)}
\end{aligned}$$

□

Problem 2. Let $q \geq 2$ be an integer and let χ be a primitive dirichlet character modulo q . Assume $\chi(-1) = 1$.

Let

$$\Lambda(s, \chi) = q^{s/2} \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) L(s, \chi)$$

Claim 2a.

$$\Lambda(s, \chi) = \epsilon \Lambda(1 - s, \bar{\chi})$$

where $\epsilon \in \mathbb{C}$ and $|\epsilon| = 1$.

Proof. $\forall s \in \mathbb{C} \ni \Re(\frac{s}{2}) > 1$ and $n \geq 1$

$$\begin{aligned}
\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) n^{-s} &= \int_0^\infty n^{-s} \pi^{-\frac{s}{2}} t^{\frac{s}{2}-1} e^{-t} dt \\
&= \int_0^\infty (n^2 \pi)^{-\frac{s}{2}} t^{\frac{s}{2}-1} e^{-t} dt && (t = n^2 \pi x) \\
&= \int_0^\infty (n^2 \pi)^{-\frac{s}{2}} (n^2 \pi x)^{\frac{s}{2}-1} e^{-n^2 \pi x} n^2 \pi dx \\
&= \int_0^\infty e^{-n^2 \pi x} x^{\frac{s}{2}-1} dx
\end{aligned}$$

Thus,

$$\begin{aligned}
\Lambda(s, \chi) &= q^{s/2} \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) L(s, \chi) \\
&= q^{\frac{s}{2}} \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \sum_{n \geq 1} \chi(n) n^{-s} \\
&= q^{\frac{s}{2}} \sum_{n \geq 1} \chi(n) \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) n^{-s} \\
&= q^{\frac{s}{2}} \sum_{n \geq 1} \chi(n) \int_0^\infty e^{-n^2 \pi x} x^{\frac{s}{2}-1} dx \\
&= q^{\frac{s}{2}} \int_0^\infty \left(\sum_{n \geq 1} \chi(n) e^{-n^2 \pi x} \right) x^{\frac{s}{2}-1} dx
\end{aligned}$$

Because of quick convergence, there is no issue with interchange of sum and integral (Tonelli Theorem and Absolute convergence).

But note

$$\begin{aligned}
\Theta(x, \chi) &= \sum_{n \in \mathbb{Z}} \chi(n) e^{in^2 \pi x} \\
&= \chi(0) + \sum_{n \geq 1} (\chi(n) + \chi(-n)) e^{in^2 \pi x} \\
&= \sum_{n \geq 1} (\chi(n) + \chi(-1)\chi(n)) e^{in^2 \pi x} \\
&= \sum_{n \geq 1} 2\chi(n) e^{in^2 \pi x} \\
&= 2 \sum_{n \geq 1} \chi(n) e^{in^2 \pi x}
\end{aligned}$$

Thus

$$\Lambda(s, \chi) = \frac{q^{\frac{s}{2}}}{2} \int_0^\infty \Theta(ix, \chi) x^{\frac{s}{2}-1} dx$$

Lets analyze the integral. We will break the integral into two parts $(0, \frac{1}{q})$ and $(\frac{1}{q}, \infty)$.

$$\int_0^\infty \Theta(ix, \chi) x^{\frac{s}{2}-1} dx = \int_0^{\frac{1}{q}} \Theta(ix, \chi) x^{\frac{s}{2}-1} dx + \int_{\frac{1}{q}}^\infty \Theta(ix, \chi) x^{\frac{s}{2}-1} dx$$

By the Jacobi Transformation Theorem,

$$\begin{aligned}
\int_0^{\frac{1}{q}} \Theta(ix, \chi) x^{\frac{s}{2}-1} dx &= \frac{\tau(\chi)}{q} \int_0^{\frac{1}{q}} \Theta\left(-\frac{1}{iq^2 x}, \bar{\chi}\right) x^{\frac{s-1}{2}} \frac{dx}{x} \\
\int_{\frac{1}{q}}^\infty \Theta(ix, \chi) x^{\frac{s}{2}-1} dx &= \frac{\tau(\chi)}{q} \int_{\frac{1}{q}}^\infty \Theta\left(-\frac{1}{iq^2 x}, \bar{\chi}\right) x^{\frac{s-1}{2}} \frac{dx}{x}
\end{aligned}$$

Let $y = \frac{1}{q^2 x}$. Then $dy = -\frac{1}{q^2 x^2} dx$. $\frac{dy}{y} = dy q^2 x = -\frac{dx}{x}$. Thus,

$$\begin{aligned} \frac{\tau(\chi)}{q} \int_0^{\frac{1}{q}} \Theta\left(-\frac{1}{iq^2 x}, \bar{\chi}\right) x^{\frac{s-1}{2}} \frac{dx}{x} &= \frac{\tau(\chi)}{q} \int_{\frac{1}{q}}^{\infty} \Theta(iy, \bar{\chi}) \left(\frac{1}{q^2 y}\right)^{\frac{s-1}{2}} \frac{dy}{y} \\ &= \frac{\tau(\chi)}{q} q^{1-s} \int_{\frac{1}{q}}^{\infty} \Theta(iy, \bar{\chi}) y^{\frac{1-s}{2}} \frac{dy}{y} \end{aligned}$$

and,

$$\begin{aligned} \frac{\tau(\chi)}{q} \int_{\frac{1}{q}}^{\infty} \Theta\left(-\frac{1}{iq^2 x}, \bar{\chi}\right) x^{\frac{s-1}{2}} \frac{dx}{x} &= \frac{\tau(\chi)}{q} \int_0^{\frac{1}{q}} \Theta(iy, \bar{\chi}) \left(\frac{1}{q^2 y}\right)^{\frac{s-1}{2}} \frac{dy}{y} \\ &= \frac{\tau(\chi)}{q} q^{1-s} \int_0^{\frac{1}{q}} \Theta(iy, \bar{\chi}) y^{\frac{1-s}{2}} \frac{dy}{y} \end{aligned}$$

Note the following

$$\begin{aligned} \tau(\chi)\tau(\bar{\chi}) &= \left(\sum_{n \in \mathbb{Z}_q} \chi(n) e\left(\frac{n}{q}\right) \right) \left(\sum_{m \in \mathbb{Z}_q} \bar{\chi}(m) e\left(\frac{m}{q}\right) \right) \\ &= \sum_{n, m \in \mathbb{Z}_q} \chi(n) \bar{\chi}(m) e\left(\frac{n+m}{q}\right) \\ &= \sum_{n, m \in \mathbb{Z}_q} \chi(nm^{-1}) e\left(\frac{n+m}{q}\right) \\ &= \sum_{a \in \mathbb{Z}_q} \chi(a) \sum_{m \in \mathbb{Z}_q} e\left(\frac{(a+1)m}{q}\right) \\ &= \sum_{a \in \mathbb{Z}_q} r\chi(a) \delta_{0, a+1} \\ &= q\chi(-1) = q \end{aligned}$$

Thus, $|\tau(\chi)| = \sqrt{q}$. Thus $|\tau(\chi)/\sqrt{q}| = 1$. Let $\epsilon = \frac{\tau(\chi)}{\sqrt{q}}$.

Thus,

$$\begin{aligned} \Lambda(s, \chi) &= \frac{\epsilon}{2} q^{\frac{1}{2}-s+\frac{s}{2}} \int_0^{\infty} \Theta(iy, \bar{\chi}) y^{\frac{1-s}{2}} \frac{dy}{y} \\ &= \frac{\epsilon}{2} q^{\frac{1-s}{2}} \int_0^{\infty} \Theta(iy, \bar{\chi}) y^{\frac{1-s}{2}} \frac{dy}{y} \end{aligned}$$

Note,

$$\Lambda(1-s, \bar{\chi}) = \frac{q^{\frac{1-s}{2}}}{2} \int_0^{\infty} \Theta(iy, \bar{\chi}) y^{\frac{1-s}{2}} \frac{dy}{y}$$

Thus,

$$\Lambda(s, \chi) = \epsilon \Lambda(1 - s, \bar{\chi})$$

□

Claim 2b. *Let $X \geq 1$ be a real number. The integral*

$$I(X, \chi) = \frac{1}{2\pi i} \int_{3-i\mathbb{R}} \Lambda(s + \frac{1}{2}, \chi) X^s \frac{ds}{s}$$

exists.

Proof. Let us examine the behavior of the integrand on the line $\Re(s) = 3$. Let $s = 3 + it$ for $t \in \mathbb{R}$.

$$\Lambda(it + \frac{7}{2}, \chi) X^{3+it} \frac{dt}{3+it}$$

Let us expand the Λ term.

$$\Lambda(it + \frac{7}{2}, \chi) = q^{\frac{it}{2} + \frac{7}{4}} \pi^{-\frac{it}{2} - \frac{7}{4}} \Gamma\left(\frac{it}{2} + \frac{7}{4}\right) L(it + \frac{7}{2}, \chi)$$

Lets start bounding Λ .

$$\begin{aligned} \left| L(it + \frac{7}{2}, \chi) \right| &\leq \sum_{n \geq 1} \left| n^{-(it + \frac{7}{2})} \right| \\ &\leq \sum_{n \geq 1} n^{-\frac{7}{2}} \\ &= \zeta\left(\frac{7}{2}\right) \end{aligned}$$

$$\begin{aligned} \left| \Gamma\left(\frac{it}{2} + \frac{7}{4}\right) \right| &= \left| \sqrt{2\pi} \left(\frac{it}{2}\right)^{\frac{5}{4}} e^{-\pi|t|/4} \left(\frac{|t|}{2e}\right)^{\frac{i}{2}t} \left(1 + O\left(\frac{1}{|t|}\right)\right) \right| \\ &= \left| \sqrt{2\pi} \left(\frac{t}{2}\right)^{\frac{5}{4}} e^{-\pi|t|/4} \left(1 + O\left(\frac{1}{|t|}\right)\right) \right| \\ &= \left| \sqrt{2\pi} \left(\frac{t}{2}\right)^{\frac{5}{4}} e^{-\pi|t|/4} + \sqrt{2\pi} \left(\frac{t}{2}\right)^{\frac{1}{4}} e^{-\pi|t|/4} \right| \\ &\leq C \left(\frac{|t|}{2}\right)^{\frac{5}{2}} e^{-\pi|t|/4} \end{aligned}$$

For the constants:

$$\left| q^{\frac{it}{2} + \frac{7}{4}} \pi^{-\frac{it}{2} - \frac{7}{4}} \right| = \left(\frac{q}{\pi}\right)^{\frac{7}{4}}$$

Thus

$$\left| \Lambda(it + \frac{7}{2}, \chi) \right| \leq C' |t|^{\frac{5}{4}} e^{-|t|/4}$$

Thus,

$$\begin{aligned} \left| \Lambda(it + \frac{7}{2}, \chi) X^{3+it} \frac{1}{3+it} \right| &= \left| \Lambda(it + \frac{7}{2}, \chi) X^{it} \frac{1}{3+it} \right| \\ &\leq C' |t|^{\frac{5}{4}} e^{-|t|/4} X^3 \frac{1}{|3+it|} \\ &\leq C' |t|^{\frac{5}{4}} e^{-|t|/4} X^3 \frac{1}{|t|} \\ &= C' |t|^{-\frac{1}{4}} e^{-|t|/4} X^3 \end{aligned}$$

as $|t| \rightarrow \infty$ $|t|^{-\frac{1}{4}} e^{-|t|/4}$ decays exponentially, ensuring the integral converges absolutely. $I(X, \chi)$ exists since it is absolutely convergent. $\square$